

LiquidAI / LFM2.5-8B-A1B

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Liquid AI

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Text Generation



Transformers



Safetensors



10 languages

lfm2_moe

liquid

lfm2.5

edge

conversational



Eval Results



arxiv:2511.23404



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Use this model



Model card



Files



xet



Community

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LFM2.5-8B-A1B

⚠ Important: The tokenizer was updated after the original release to fix tool-calling issues in `llama.cpp`. If you downloaded LFM2.5-8B-A1B before [commit feb5e04](#), please re-download the tokenizer files. The [GGUF files](#) have also been re-converted with the updated tokenizer.

LFM2.5 is a new family of hybrid models designed for on-device deployment. It builds on the LFM2 architecture with extended pre-training and reinforcement learning.

- On-device personal assistant:** Designed to power real-life applications, chaining tool calls, and following complex instructions on all devices.

Downloads last month

175,137



Safetensors

Model size

8B params

Tensor type

F32 · BF16

Chat template

Files info

Inference Providers

Text Generation

This model isn't deployed by any Inference Provider.



8 Ask for provider support

Model tree for LiquidAI/LFM2.5-8...

Base model LiquidAI/LFM2.5-8B-A1B-...



Finetuned (29)

[this model](#)

Adapters

28 models

Finetunes

30 models

Quantizations

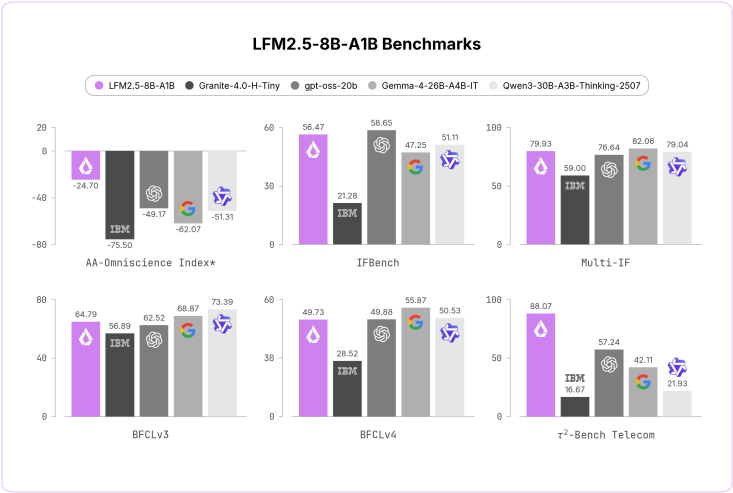


55 models

Spaces using LiquidAI/LFM2.5-... 8

- **Compressed performance:** Competitive with much larger dense and MoE models on instruction following and agentic tasks.
- **Unmatched throughput:** Fastest in its size class on both CPU and GPU inference, with day-one support for llama.cpp, MLX, vLLM, and SGLang.

Find more information about LFM2.5-8B-A1B in our [blog post](#).



*AA-Omniscience Index (higher is better) rewards correct answers and penalizes hallucinations. Scores range from -100 to 100. See more results on [Artificial Analysis](#).

Model Details

Model	Parameters	Description
LFM2.5-8B-A1B-Base	8.3B total / 1.5B active	Pre-trained base model for fine-tuning
LFM2.5-8B-A1B	8.3B total / 1.5B active	Reasoning-tuned general-purpose model

- 🔥 [LiquidAI/LFM2.5-8B-A1B](#)
- 🔗 [usermma/LFM2.5-8B-A1B-Abliterated...](#)
- 🔗 [akhaliq/LFM2.5-8B-A1B](#)
- 🔗 [Maharnaveed/LFM2.5-8B-A1B](#)
- 🔗 [chmielvu/LFM2.5-8B-A1B-Abliterated...](#)
- 🔗 [developerjeremylive/LFM2.5-8B-A1B...](#)
- 🔗 [MhmodijlAi/LFM2.5-8B-A1B-Abliterate...](#)
- 🔗 [tenet/LFM2.5-8B-A1B](#)

Collection including LiquidAI/LFM...

💧 **LFM2.5** 📁 Collection

Collection ... • 14 items • Up... • ⬆ 168

Paper for LiquidAI/LFM2.5-8B-A1B

LFM2 Technical Report

📄 Paper • 2511.23404 • Publi... • ⬆ 65

📊 Evaluation results ⓘ

MathArena/aime_2026 • MathA... 📄 ➦ 50

LFM2.5-8B-A1B is a general-purpose text-only model with the following features:

- **Total parameters:** 8.3B
- **Active parameters:** 1.5B
- **Number of layers:** 24 (18 double-gated LIV conv + 6 GQA)
- **Training budget:** 38 trillion tokens
- **Context length:** 128,000
- **Vocabulary size:** 128,000
- **Languages:** English, Arabic, Chinese, French, German, Italian, Japanese, Korean, Portuguese, Spanish
- **Generation parameters:** We recommend the following parameters:
 - temperature: 0.2
 - top_k: 80
 - repetition_penalty: 1.05

Model	Description
LFM2.5-8B-A1B	Original model checkpoint in native format. Best for fine-tuning or inference with Transformers, vLLM, and SGLang.
LFM2.5-8B-A1B-GGUF	Quantized format for llama.cpp and compatible tools. Optimized for edge inference and local deployment.
LFM2.5-8B-A1B-ONNX	ONNX Runtime format for cross-platform deployment.
LFM2.5-8B-A1B-MLX	MLX format for Apple Silicon. Optimized for fast inference on Mac devices.

We recommend using LFM2.5-8B-A1B for agentic workflows, tool use, structured outputs, multilingual assistants, and on-device personal-assistant applications. It is not the best fit for heavy programming or knowledge-intensive question answering without retrieval.

Chat Template

LFM2.5 uses a ChatML-like format. See the [Chat Template documentation](#) for details. Example:

```
<|startoftext|><|im_start|>system
You are a helpful assistant trained by Liquid
<|im_start|>user
What is C. elegans?<|im_end|>
<|im_start|>assistant
```

Because LFM2.5-8B-A1B is a reasoning model, assistant turns contain an explicit chain of thought before the final answer. You can use `tokenizer.apply_chat_template()` to format your messages automatically.

Tool Use

LFM2.5 supports function calling in four steps:

1. **Function definition:** Provide the list of tools as a JSON object in the system prompt, or use `tokenizer.apply_chat_template()` with `tools=....`
2. **Function call:** By default, LFM2.5 writes Pythonic function calls (a Python list between `<|tool_call_start|>` and `<|tool_call_end|>` special tokens), as the assistant answer. You

can override this behavior by asking the model to output JSON function calls in the system prompt.

- 3. **Function execution:** Execute the call and return the result with the `tool` role.
- 4. **Final answer:** LFM2.5 interprets the tool output and returns a plain-text answer addressing the original prompt.

See the [Tool Use documentation](#) for the full guide.

Example:

```
<|startoftext|><|im_start|>system
List of tools: [{"name": "get_candidate_status"}]
<|im_start|>user
What is the current status of candidate ID 12345?
<|im_start|>assistant
<|tool_call_start|>[get_candidate_status(candidate_id="12345")]
<|im_start|>tool
[{"candidate_id": "12345", "status": "Interview scheduled"}]
<|im_start|>assistant
The candidate with ID 12345 is currently in the interview stage.
```

 Inference

LFM2.5-8B-A1B is supported by many inference frameworks. See the [Inference documentation](#) for the full list.

Name	Description	Docs	Notebook
Transformers	Simple inference with direct access to model internals.	Link	 Open in Colab
vLLM	High-throughput production	Link	 Open in Colab

Name	Description	Docs	Notebook
	deployments with GPU.		
llama.cpp	Cross-platform inference with CPU offloading.	Link	
MLX	Apple's machine learning framework optimized for Apple Silicon.	Link	—
LM Studio	Desktop application for running LLMs locally.	Link	—

Quick start with Transformers (compatible with `transformers>=5.0.0`):

```

from transformers import AutoModelForCausalLM

model_id = "LiquidAI/LFM2.5-8B-A1B"
model = AutoModelForCausalLM.from_pretrained(
    model_id,
    device_map="auto",
    dtype="bfloat16",
    # attn_implementation="flash_attention_2"
)

tokenizer = AutoTokenizer.from_pretrained(model_id)
streamer = TextStreamer(tokenizer, skip_prompt=True)

prompt = "What is C. elegans?"

input_ids = tokenizer.apply_chat_template(
    [{"role": "user", "content": prompt}],
    add_generation_prompt=True,
    return_tensors="pt",
    tokenize=True,

```

```
)["input_ids"].to(model.device)

output = model.generate(
    input_ids,
    do_sample=True,
    temperature=0.2,
    top_k=80,
    repetition_penalty=1.05,
    max_new_tokens=8192,
    streamer=streamer,
)
```

 Fine-Tuning

We recommend fine-tuning LFM2.5 for your specific use case to achieve the best results.

Name	Description	Docs	Notebook
CPT (Unsloth)	Continued Pre-Training using Unsloth for text completion.	Link	 Open in Colab
CPT (Unsloth)	Continued Pre-Training using Unsloth for translation.	Link	 Open in Colab
SFT (Unsloth)	Supervised Fine-Tuning with LoRA using Unsloth.	Link	 Open in Colab
SFT (TRL)	Supervised Fine-Tuning with LoRA using TRL.	Link	 Open in Colab
DPO (TRL)	Direct Preference Optimization with LoRA using TRL.	Link	 Open in Colab

Name	Description	Docs	Notebook
GRPO (Unsloth)	GRPO with LoRA using Unsloth.	Link	 Open in Colab
GRPO (TRL)	GRPO with LoRA using TRL.	Link	 Open in Colab



Performance

Improvements over LFM2-8B-A1B

Thanks to reasoning, scaled-up pre-training, and large-scale RL, LFM2.5-8B-A1B improves over its predecessor across the board:

Benchmark	LFM2-8B-A1B	LFM2.5-8B-A1B	Δ
AA-Omniscience Index	-78.42	-24.70	+53.62
AA-Omniscience Accuracy	7.33	8.67	+1.34
AA-Omniscience Non-Hallucination Rate	7.46	63.47	+56.01
IFEval	79.44	91.84	+12.40
IFBench	26.00	56.47	+30.47
Multi-IF	58.54	79.93	+21.39
MATH500	74.80	88.76	+13.96
AIME25	20.00	42.53	+22.53
BFCLv3	45.07	64.36	+19.29
BFCLv4	25.52	48.50	+22.98

Benchmark	LFM2-8B-A1B	LFM2.5-8B-A1B	Δ
Tau ² Telecom	13.60	88.07	+74.47
Tau ² Retail	7.02	39.82	+32.80

Knowledge and instruction following

Model	Parameters	AA-Omni. Index	AA-Omni. Accuracy	AA-Omni. Non-Halluc.	I
LFM2.5-8B-A1B	8B/A1B	-24.70	8.67	63.47	
Granite-4.0-H-Tiny	7B/A1B	-75.50	9.37	6.38	
Qwen3.5-4B	4B	-51.53	17.20	16.99	
Qwen3-30B-A3B-Thinking-2507	30.5B/3.3B	-51.31	18.80	13.87	
Gemma-4-E2B-IT	5.1B	-72	7.00	15.05	
Gemma-4-E4B-IT	8B	-50.67	8.10	36.06	
Gemma-4-26B-A4B-IT	26B/4B	-62.07	14.37	10.75	
gpt-oss-20b	21B/3.6B	-49.17	14.57	24.50	

Math and agentic workflows

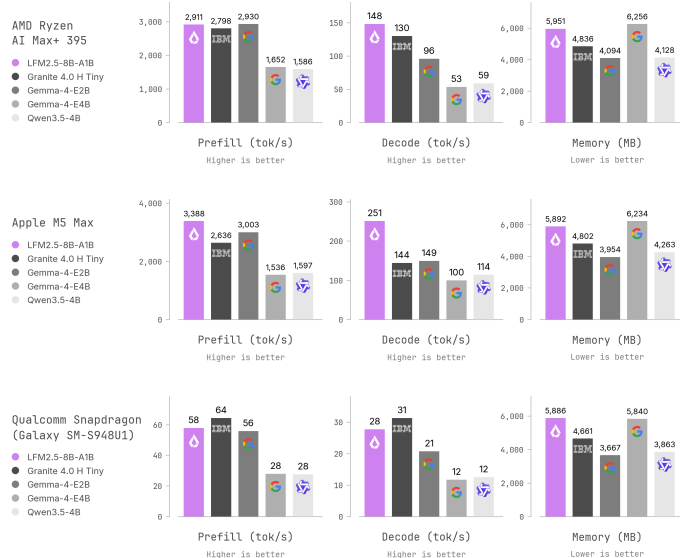
Model	Parameters	MATH500	AIME25	AIME26
LFM2.5-8B-A1B	8B/A1B	88.76	42.53	50.00
Granite-4.0-H-Tiny	7B/A1B	59.20	4.93	3.33
Qwen3.5-4B	4B	80.76	54.28	58.33
Qwen3-30B-A3B-Thinking-2507	30.5B/3.3B	86.48	71.67	66.67
Gemma-4-E2B-IT	5.1B	64.00	26	30
Gemma-4-E4B-IT	8B	65.00	34.33	40.67

CPU Inference

LFM2.5-8B-A1B

CPU Inference Metrics by Device

Q4_K_M quantization and 4K-token input context



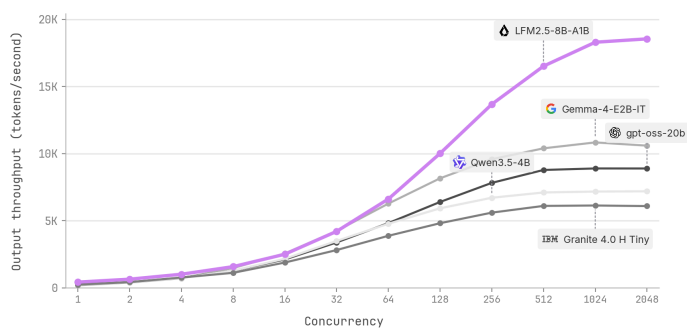
GPU Inference

LFM2.5-8B-A1B is the fastest model in its size class, reaching **18.5K output tokens per second at high concurrency**, over 1.6B tokens per day on a single H100.

LFM2.5-8B-A1B

GPU Inference Throughput by Concurrency

SGLang | Input: 1024 tokens | Output: 256 tokens



Contact

- Got questions or want to connect? [Join our Discord community](#).
- If you are interested in custom solutions with edge deployment, please contact [our sales](#)

team.

Citation

System theme TOS Privacy About Careers 🤖 Models Datasets Spaces Pricing Docs

```
@article{liquidai20268BA1B,  
  title = {LFM2.5-8B-A1B: Personal Assistant},  
  author = {Liquid AI},  
  year = {2026},  
  note = {www.liquid.ai/blog/lfm2-5-8b-a1b}  
}
```

```
@article{liquidai2025lfm2,  
  title = {LFM2 Technical Report},  
  author = {Liquid AI},  
  journal = {arXiv preprint arXiv:2511.23404},  
  year = {2025}  
}
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